PATUAKHALI SCIENCE AND TECHNOLOGY UNIVERSITY

COURSE CODE EEE-212

Electrical Technology Sessional

Project Report: Smart Plant Monitoring System Using NodeMCU ESP8266

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1. Introduction

1.1 Background

As urbanization increases, many people find it challenging to maintain healthy plants due to busy lifestyles and environmental factors. This project aims to develop a Smart Plant Monitoring System that utilizes IoT technology to automate plant care. By continuously monitoring soil moisture, temperature, and humidity, this system can help users maintain optimal growing conditions for their plants.

1.2 Objectives

- To design a system that monitors environmental conditions affecting plant health.
- To provide real-time data to users through a mobile application (Blynk).
- To automate the watering process based on soil moisture levels.
- To implement a security feature using a PIR motion sensor.

2. Materials and Components

- **Microcontroller:** NodeMCU ESP8266
- **Sensors:**
    - Soil Moisture Sensor
    - DHT11 Temperature and Humidity Sensor
    - PIR Motion Sensor
- **Modules:**
    - Relay Module for automatic watering
    - I2C Liquid Crystal Display (LCD)
- **Miscellaneous:**
    - Push Button for manual control
    - Jumper wires and breadboard
    - Power supply (USB or battery)

3. System Architecture

3.1 Hardware Architecture

The system architecture comprises the following components:

- The NodeMCU microcontroller connects to various sensors and modules.
- The DHT11 sensor measures temperature and humidity.
- The soil moisture sensor detects moisture levels in the soil.
- The PIR sensor detects motion for security.
- The relay controls the water pump, activating it based on soil moisture readings.
- An LCD displays real-time data for immediate feedback.

3.2 Software Architecture

The software architecture includes:

- **Programming Language:** Arduino IDE for writing and uploading code to the NodeMCU.
- **Libraries Used:**
    - LiquidCrystal_I2C for LCD control
    - ESP8266WiFi and BlynkSimpleEsp8266 for IoT connectivity
    - DHT for temperature and humidity readings

4. Implementation

4.1 Circuit Design

The components were connected as follows:

Component	NodeMCU Pin
Soil Moisture Sensor	A0
PIR Motion Sensor	D6
Relay Module	D7
Push Button	D5
DHT Sensor (DHT11)	D4
LCD (I2C)	SDA (D2), SCL (D1)

4.2 Software Development

The code was structured to initialize sensors, read their values, and send the data to the Blynk app. Key functionalities included:

- **Sensor Readings:** Periodic updates for temperature, humidity, and soil moisture.
- **Relay Control:** Automated watering based on soil moisture levels and manual control via a push button or the Blynk app.
- **User Interface:** Real-time data displayed on the LCD and Blynk app notifications for sensor alerts.

4.3 Code Overview

The main functionalities included:

- Initialization of components in the `setup()` function.
- Continuous monitoring of sensors in the `loop()` function, with data updates every few seconds.

5. Results

5.1 Data Collection

- The DHT11 sensor consistently provided accurate temperature and humidity readings.

- Soil moisture levels were displayed as percentages, with readings effectively communicating the need for watering.
- The PIR sensor successfully detected motion, triggering alerts in the Blynk app.

5.2 System Performance

The system operated effectively, maintaining real-time monitoring of environmental conditions. Users were able to receive notifications and manage watering from the Blynk app, enhancing the convenience of plant care.

Observations and Data

Parameter	Measurement	Threshold/Condition
Soil Moisture	45%	Water pump ON when <30%
Temperature	25°C	Displayed on LCD and Blynk
Humidity	60%	Displayed on LCD and Blynk
PIR Motion Detected	Yes	Notification on Blynk
Manual Water Control	Functional	Via push button and Blynk

6. Discussion

6.1 Challenges

- **Connectivity Issues:** Initially, there were challenges with Wi-Fi connectivity, which were resolved by relocating the NodeMCU closer to the router.
- **Sensor Calibration:** The soil moisture sensor required calibration to provide accurate readings across different soil types.

6.2 Future Improvements

- **Enhanced User Interface:** Developing a web interface for broader accessibility.
- **Integration of Additional Sensors:** Adding light sensors or nutrient sensors for a comprehensive monitoring system.
- **Machine Learning Algorithms:** Implementing predictive analytics for watering schedules based on historical data.

7. Conclusion

The Smart Plant Monitoring System successfully integrates IoT technology with environmental monitoring to automate plant care. By providing real-time data and remote control capabilities, the system improves the efficiency of plant management. This project has significant potential for further development in smart agriculture and home gardening.